

Chapter 7 Assignment — Analytic Trigonometry

Name: _____

Date: _____

How to use this set: Each topic begins with *Strategy Notes* and a *Worked Example*. Then try the *Practice* (skill) and the *Challenge* (deeper) items. Give exact answers using radicals and π when possible. State domain restrictions when solving equations/inequalities.

1. Topic 1: Core Trigonometric Identities

Strategy Notes. Use the Pythagorean identities $\sin^2 x + \cos^2 x = 1$, $1 + \tan^2 x = \sec^2 x$, and $1 + \cot^2 x = \csc^2 x$. Convert everything to sines and cosines when stuck, factor common terms, and avoid cross-multiplying until both sides are simplified.

Worked Example. Verify the identity $\tan \theta \sin \theta + \cos \theta = \sec \theta$.

Solution. $\tan \theta \sin \theta = \frac{\sin \theta}{\cos \theta} \sin \theta = \frac{\sin^2 \theta}{\cos \theta}$. Then $\frac{\sin^2 \theta}{\cos \theta} + \cos \theta = \frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta} = \frac{1}{\cos \theta} = \sec \theta$.

Practice.

- (a) Prove the identity $\sin x \tan\left(\frac{x}{2}\right) = 1 - \cos x$.

(b) Show that $\cos 4x = 1 - 8 \sin^2 x + 8 \sin^4 x$.

Challenge.

(a) Verify $\frac{\cot x - 1}{\cot x + 1} = \frac{\cos 2x}{1 + \sin 2x}$.

2. Topic 2: Addition and Subtraction Formulas

Strategy Notes. $\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$ and $\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$. Use these to rewrite linear combinations of $\sin x$ and $\cos x$, and to evaluate nonstandard angles.

Worked Example. Simplify $\sin(x - y) \cos y + \cos(x - y) \sin y$.

Solution. By the sine addition formula, this equals $\sin((x - y) + y) = \sin x$.

Practice.

(a) Rewrite $\sqrt{2} \cos\left(\frac{\pi}{4} - x\right)$ in the form $A \cos x + B \sin x$ and give A, B .

(b) Express $\cos(\theta - \pi)$ entirely in terms of $\sin \theta$ and/or $\cos \theta$.

Challenge.

- (a) Evaluate $\tan 285^\circ$ exactly.

3. Topic 3: Double-, Half-Angle, and Product-to-Sum Formulas

Strategy Notes. Use $\sin 2x = 2 \sin x \cos x$, $\cos 2x = \cos^2 x - \sin^2 x = 1 - 2 \sin^2 x = 2 \cos^2 x - 1$. Half-angle: $\sin^2\left(\frac{x}{2}\right) = \frac{1 - \cos x}{2}$ and $\cos^2\left(\frac{x}{2}\right) = \frac{1 + \cos x}{2}$. Product-to-sum converts products like $\sin ax \sin bx$ to sums.

Worked Example. Compute $\sin\left(\frac{\pi}{12}\right)$ exactly.

Solution. $\frac{\pi}{12} = \frac{\pi}{3} - \frac{\pi}{4}$. Then $\sin\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = \sin \frac{\pi}{3} \cos \frac{\pi}{4} - \cos \frac{\pi}{3} \sin \frac{\pi}{4} = \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} - \frac{1}{2} \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{6} - \sqrt{2}}{4}$.

Practice.

(a) If $\sin 2x = \frac{24}{25}$ with $0 < x < \frac{\pi}{2}$, find $\sin^4 x + \cos^4 x$.

(b) Find the exact value of $\sin 75^\circ$.

Challenge.

- (a) Determine constants a, b so that $\frac{-8 + 9 \cos^2 \theta}{1 - 3 \sin \theta} = a + b \sin \theta$.

4. Topic 4: Solving Trigonometric Equations and Inequalities

Strategy Notes. Isolate a basic trig function, use identities to factor/rewrite, and check that solutions lie in the stated interval. For inequalities, analyze sign charts or transform to equivalent algebraic inequalities on reference angles.

Worked Example. Solve $\sin 2x = \sin x$ for $0 \leq x \leq 2\pi$.

Solution. $\sin 2x - \sin x = \sin x(2 \cos x - 1) = 0$. Thus $\sin x = 0$ gives $x = 0, \pi, 2\pi$, and $2 \cos x - 1 = 0$ gives $\cos x = \frac{1}{2}$, i.e., $x = \frac{\pi}{3}, \frac{5\pi}{3}$.

Practice.

(a) Find all x with $0 \leq x \leq 2\pi$ such that $\sin 2x \leq \sin x$.

(b) For $-\pi \leq \theta \leq \pi$, solve $2(\sin^2 \theta - \cos^2 \theta) = 8 \sin \theta - 5$.

Challenge.

- (a) Solve $\cos 3x = \sin x$ for $0 \leq x < 2\pi$.

5. Topic 5: Phase-Amplitude Form and Mixed Skills

Strategy Notes. Any expression of the form $A \sin t + B \cos t$ can be written as $R \sin(t + \varphi)$ where $R = \sqrt{A^2 + B^2}$ and $\tan \varphi = \frac{B}{A}$. This makes ranges and extrema immediate.

Worked Example. Find the range of $g(t) = 2 \sin t + 3 \cos t$.

Solution. Write $2 \sin t + 3 \cos t = R \sin(t + \varphi)$ with $R = \sqrt{2^2 + 3^2} = \sqrt{13}$. Hence $-\sqrt{13} \leq g(t) \leq \sqrt{13}$.

Practice.

(a) If $\cos \theta = -0.6$, find all possible values of $\sin 2\theta$.

(b) If $\sin x + \cos x = \frac{1}{5}$ and $\frac{\pi}{2} < x < \pi$, find $\tan x$.

Challenge.

- (a) Suppose $\sin a + \sin b = \sqrt{\frac{5}{3}}$ and $\cos a + \cos b = 1$. Find $\cos(a - b)$.